Incident report: fraud-scorer feature store lag

Document: INC-403

Date: 2026-04-22

Services: fraud-scorer, payments-api

Summary

SEV3 incident on April 22, 2026 affecting fraud-scorer and payments-api. The fraud-scorer feature store fell behind with a lag of 12 minutes, causing payment approvals to wait on fresh features and become slower. Marco Bellini investigated. The issue was resolved by restarting the feature consumer.

Impact

Payment approval latency increased by 3 seconds slower during the incident window. Merchants experienced slower payment processing as the fraud-scorer service waited for feature store updates. See STS-403 for real-time status updates during the incident.

Timeline

- 14:32 UTC: Monitoring alert triggered for feature store lag
- 14:35 UTC: Incident commander notified; Marco Bellini began investigation
- 14:41 UTC: Root cause identified as feature consumer process stalled
- 14:43 UTC: Feature consumer restarted
- 14:45 UTC: Feature store lag returned to normal; payment approval latency normalized

Root cause

The fraud-scorer feature store consumer process stopped consuming messages from the upstream data pipeline, causing the feature store to fall behind by 12 minutes. This lag forced the payments-api to wait for fresh features during fraud scoring, adding latency to payment approvals.

Resolution

The feature consumer was restarted, which resumed normal message consumption and cleared the accumulated lag. Payment approval latency returned to baseline within 2 minutes of restart.

Follow-up actions

- Implement automated alerting for feature store lag as described in CHG-1080
- Review RB-12 for standard procedures to follow during future feature store lag incidents
- Investigate why the feature consumer process stalled and implement safeguards to prevent recurrence